DIGITAL FORENSICS PRACTICAL

Disk Imaging and Deleted File Recovery

Using FTK Imager and Autopsy

Case FTK-2026-001 · Evidence E-001 · Examiner: Abdul Ghafoor

Objective and Tools

Objective

Acquire a forensic image of a drive, verify its integrity, then analyze it to locate and recover deleted files.

Tools

FTK Imager 8.3 for acquisition and verification. Autopsy 4.23.1 for analysis and recovery.

Setup

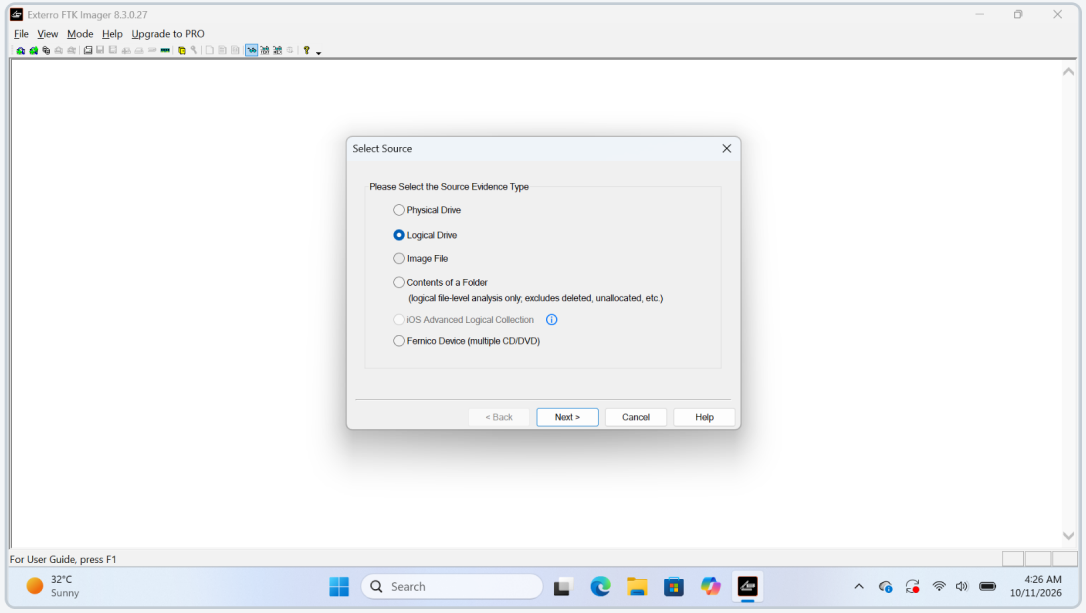
Source: D:\ (NTFS, logical drive). Destination: E:\ as a Raw (dd) image of about 24.5 GB.

Workflow Overview

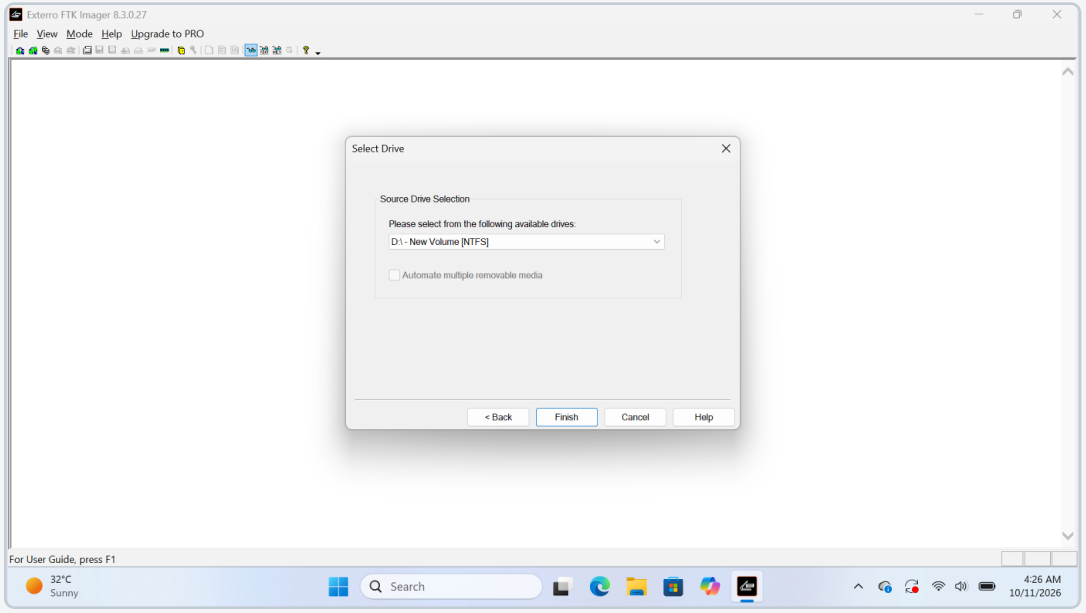
1. Acquire
2. Verify
3. Create case
4. Analyze
5. Recover

Steps 1 and 2 are done in FTK Imager (image creation, MD5 and SHA1 hashing). Steps 3 to 5 are done in Autopsy (case setup, ingest of the image, review of deleted files and recovery).

FTK Imager: Select Source and Drive

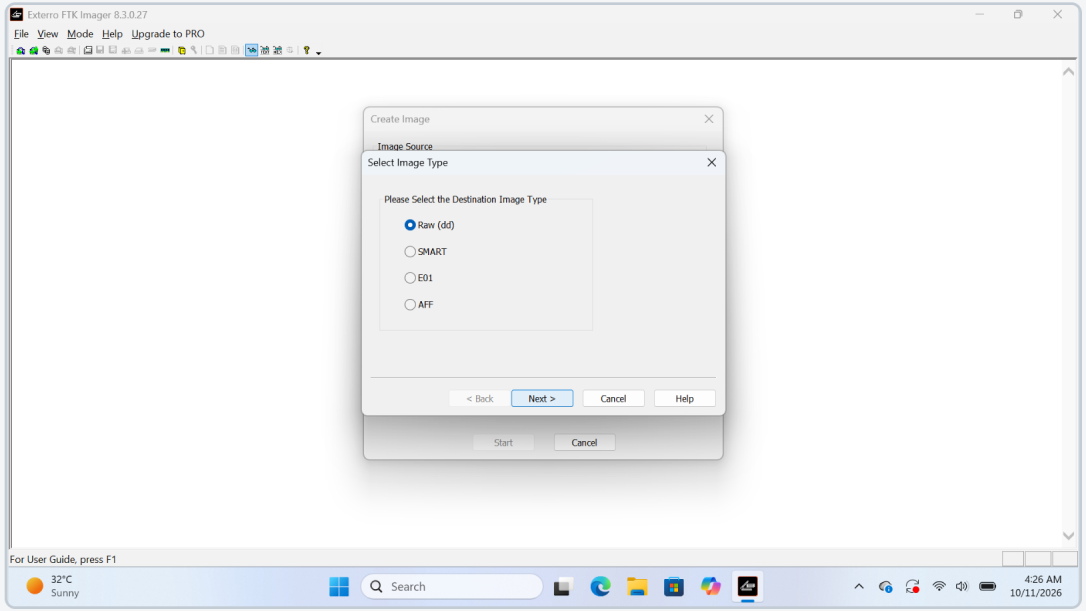


Choose Logical Drive as the evidence type.

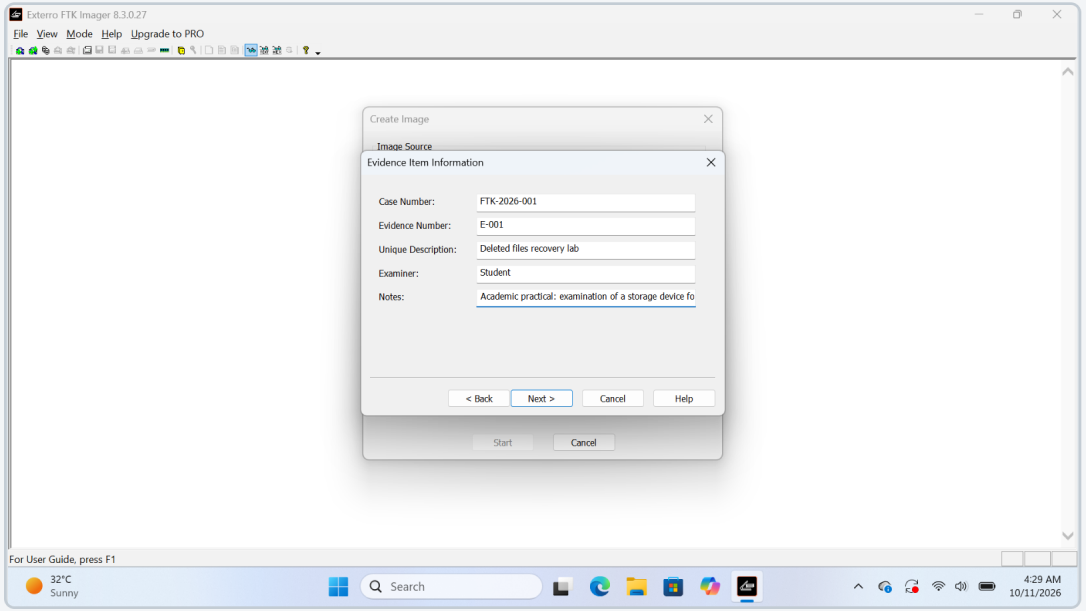


Select D:\ (New Volume, NTFS) as the source.

FTK Imager: Image Type and Evidence Details

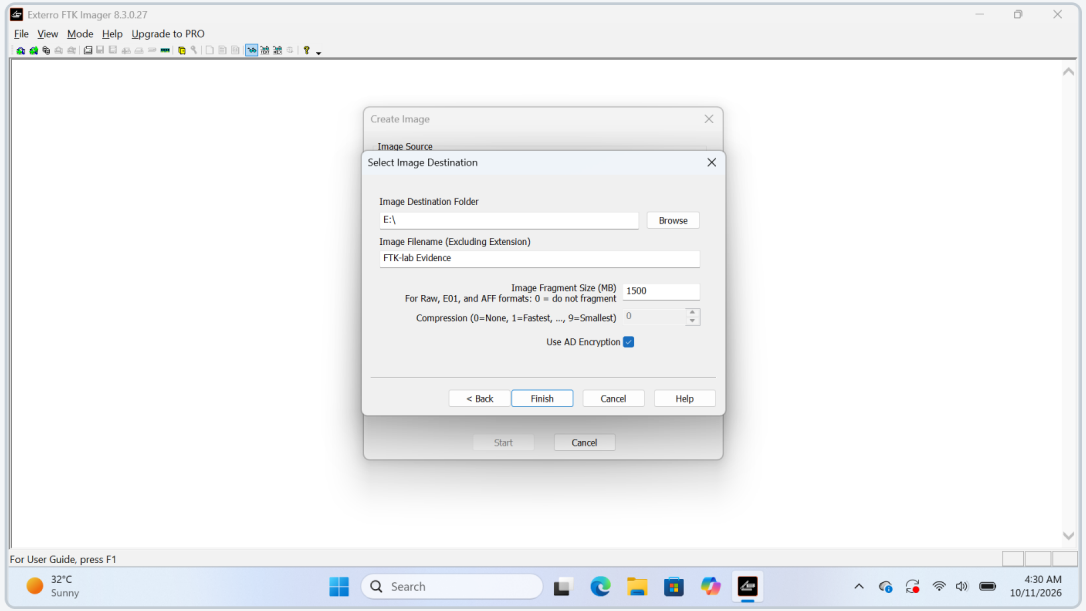


Destination image type: Raw (dd).

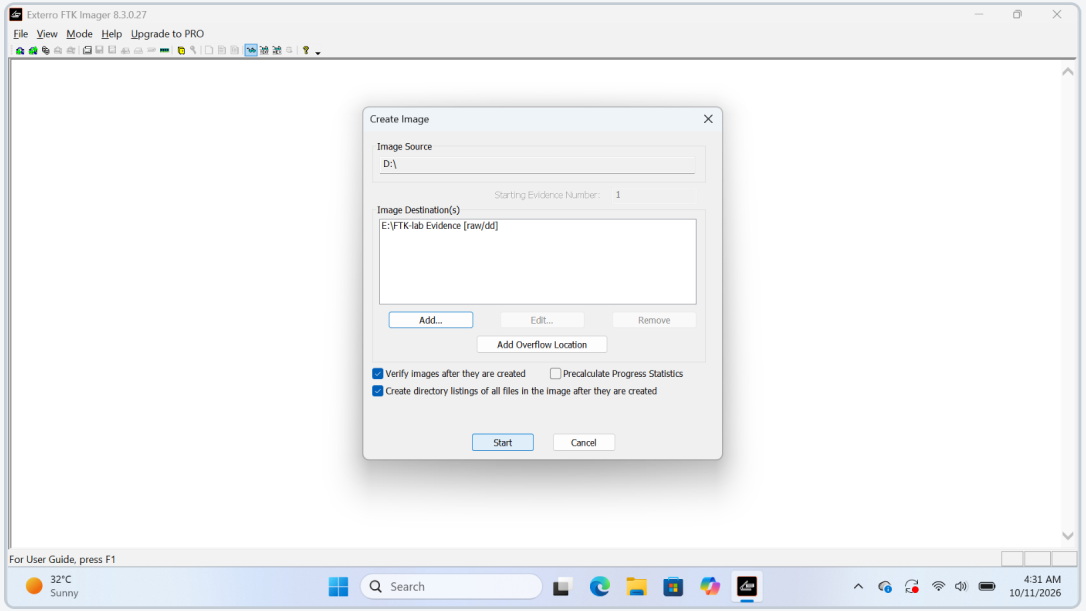


Case FTK-2026-001, evidence E-001, examiner and notes.

FTK Imager: Destination and Options



Destination E:\, filename FTK-lab Evidence, 1500 MB fragments.



Verify image and directory listing options enabled.

FTK Imager: Image Created and Verified

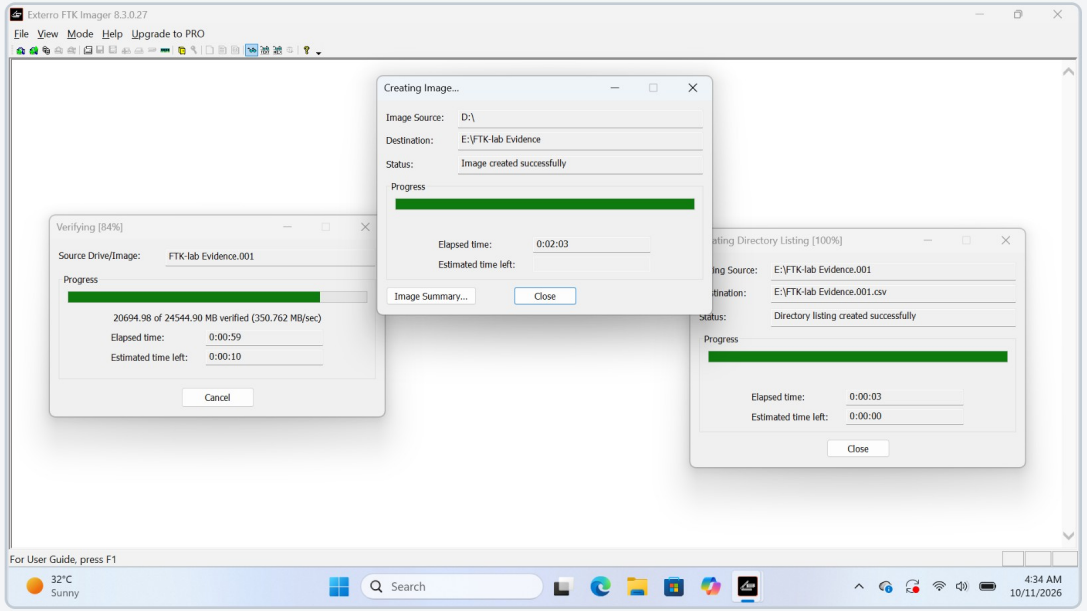
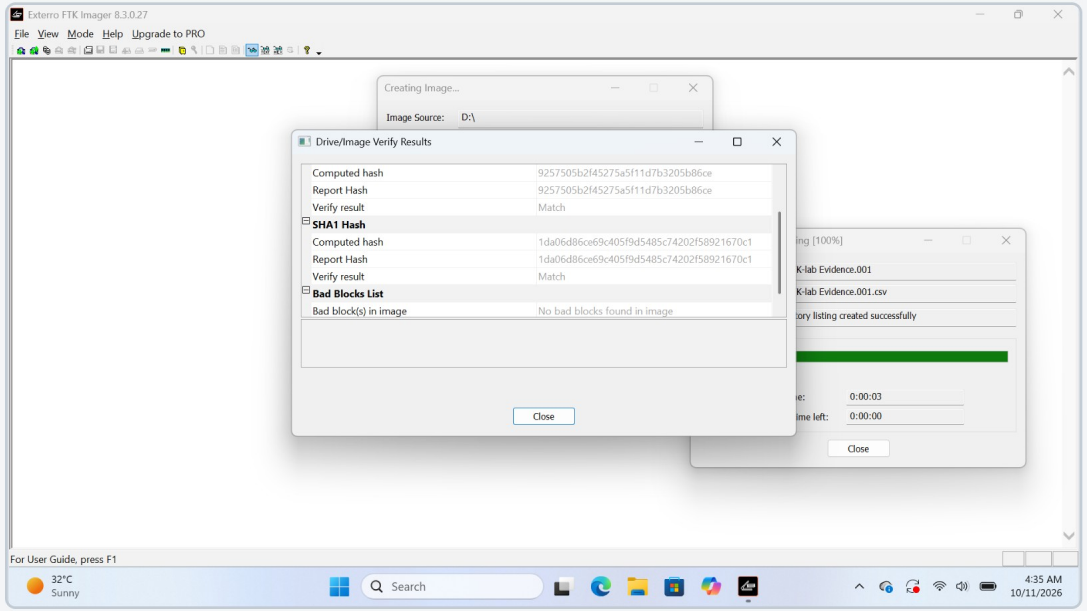
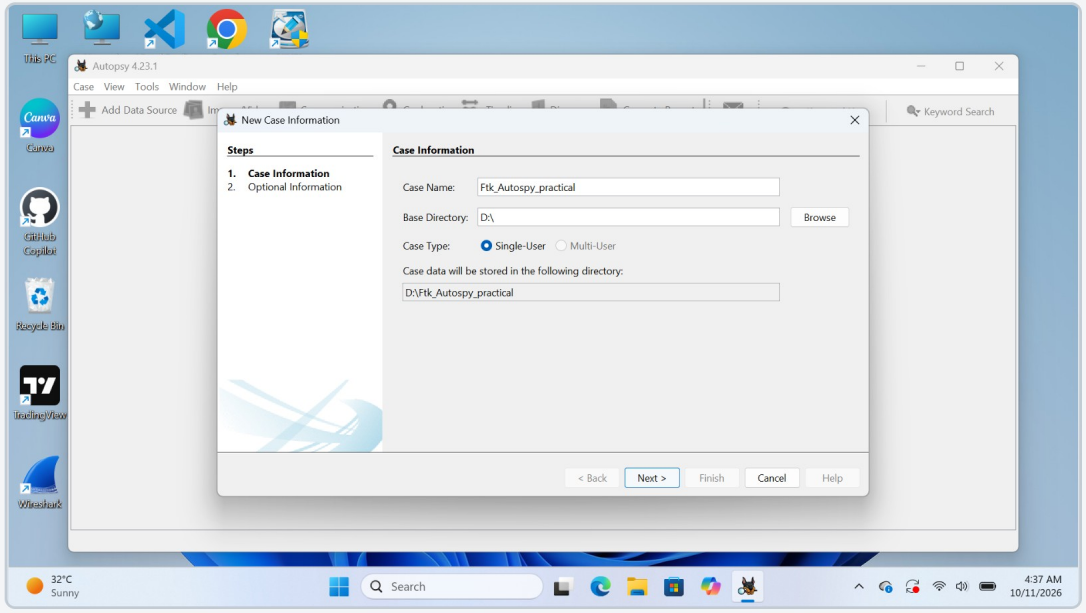


Image created successfully in 2 min 03 s.

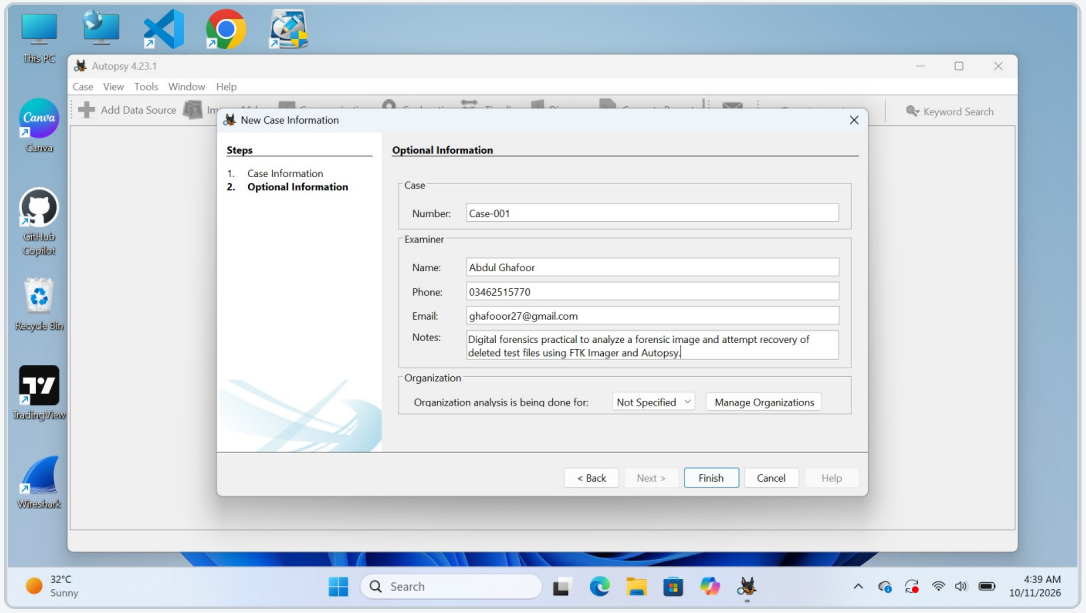


Computed and report hashes match; no bad blocks.

Autopsy: Create the Case



New case Ftk_Autopsy_practical, single-user, stored on D:\.



Case number, examiner details and case notes.

Autopsy: Add the Image as Data Source

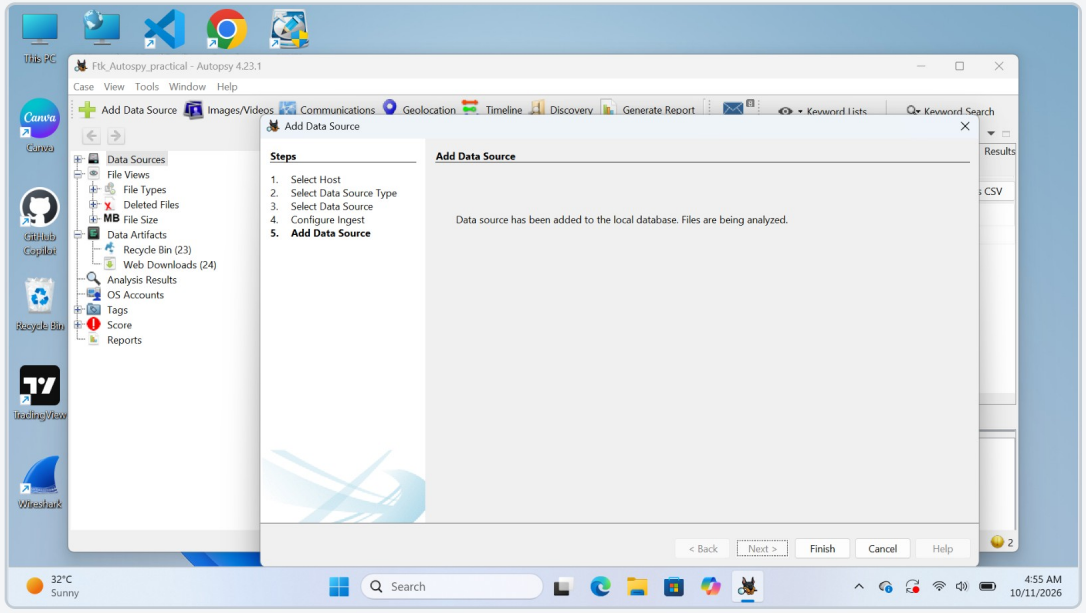


Image added; files analyzed by ingest modules.

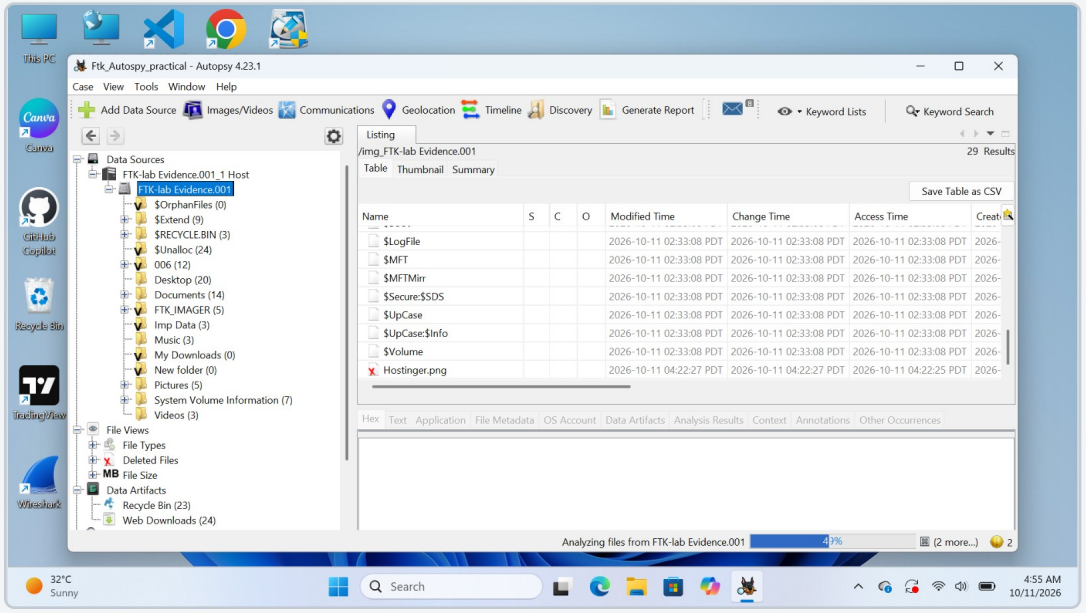
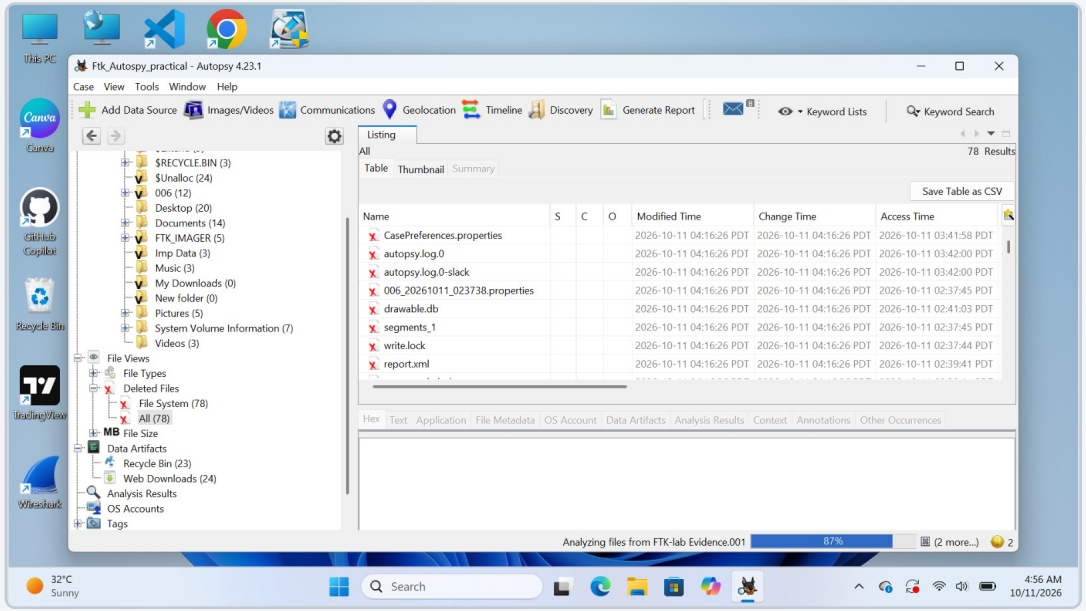
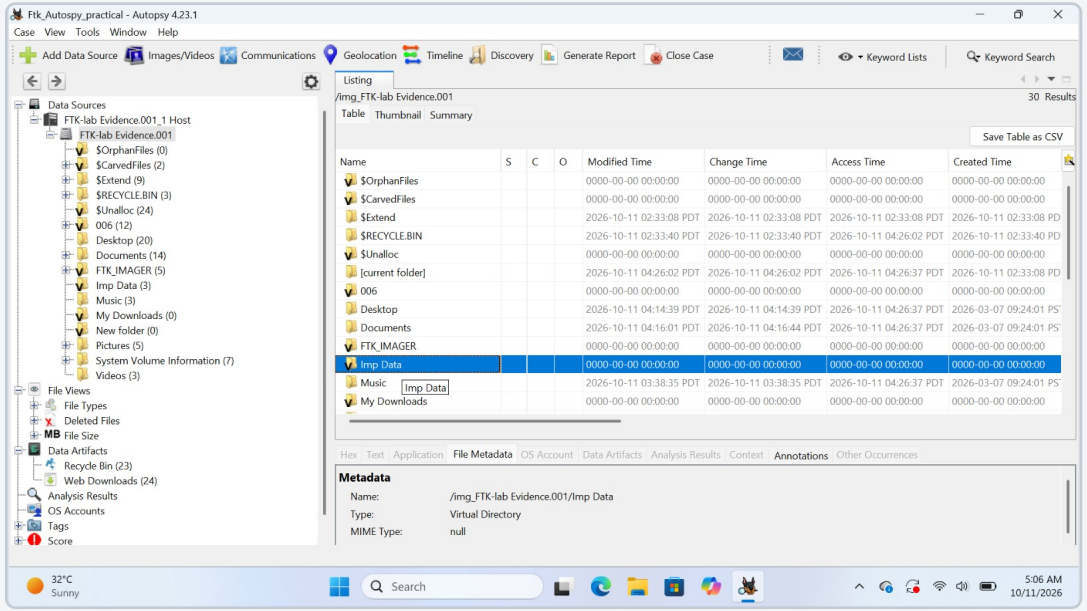


Image file system tree with folders and files listed.

Autopsy: Locate Deleted Files

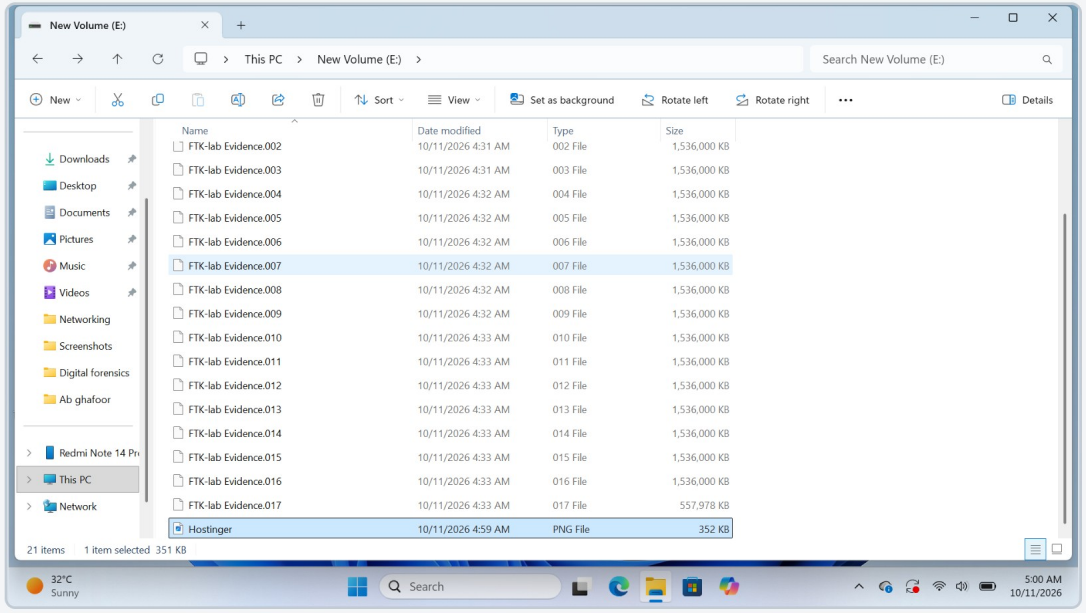


Deleted Files view: 78 file-system results.

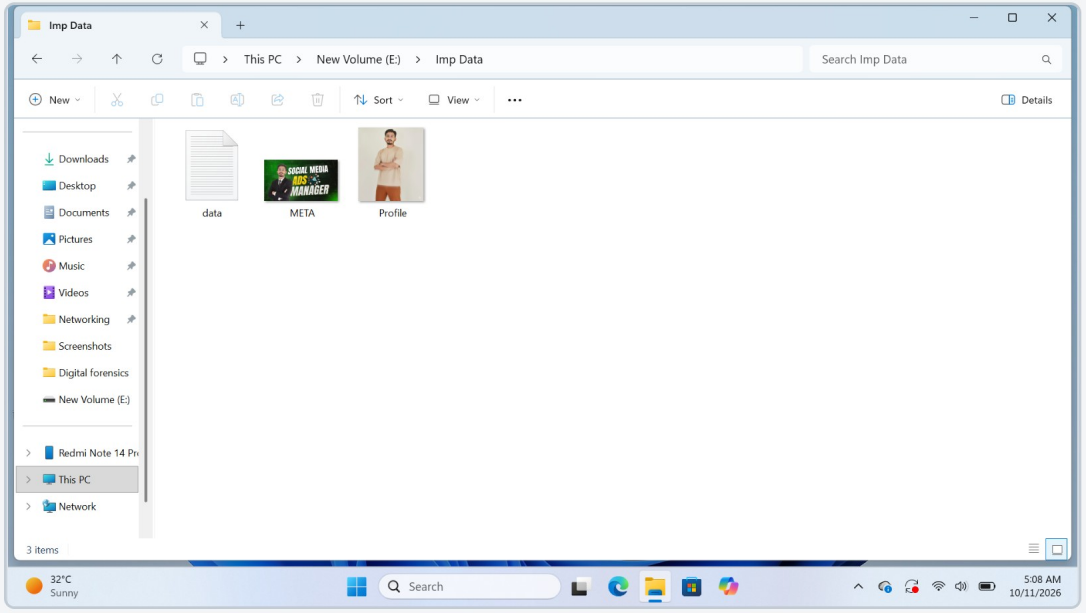


Imp Data folder located inside the image.

Recovery Results



Deleted file exported to E:\ and confirmed in File Explorer.



Imp Data folder recovered: data, META and Profile.

Conclusion

Acquired

24,544 MB drive imaged as Raw (dd) with FTK Imager.

Verified

MD5 and SHA1 hashes matched, with zero bad blocks.

Recovered

Autopsy exposed deleted files and the Imp Data folder was recovered.

The practical shows the standard forensic workflow: acquire, verify integrity, analyze and recover.